DAY 07 / 11 · WEB DEV IN 11 DAYS · GUIDED

Make the page react

The DOM and events

ShowShelf after today: **cards built by JavaScript, live search and a My List that survives a reload**

TODAY

The plan for the next 90 minutes

Minutes	Topic
10	Recap: five questions about Day 6
8	The DOM: find an element
10	Build an element
12	Data in, elements out
12	Events
10	State → render
8	My List and localStorage
5	The big idea, and questions
15	This afternoon in the Lab

RECAP · DAY 6

Five quick questions

1. favourites[0] and favourites.length: what do they give?
2. Read shows[1].rating aloud. What is it?
3. filter, map or find: all names? rated 9 or more? id 2?
4. Why does topRated(list) use list, not shows?
5. How do you check whether a show is a comedy?

THE DOM

Find an element, then change it

The browser turns your HTML into objects: the **DOM**. Every tag is an **element**.

querySelector takes a CSS selector: **#** for an id, **.** for a class.

```
const heading = document.querySelector("h1");
heading.textContent = "Movie night!";
console.log(heading);
```

Found nothing? You get null, then a red error: Cannot set properties of null. Check the # or the . first.

BUILD AN ELEMENT

Create it, fill it, put it on the page

```
const note = document.createElement("p");        // 1. create (not on the page yet)
note.className = "status";                       // 2. give it a class...
note.textContent = "12 shows tonight";           //    ...and some text
document.querySelector(".hero").append(note);    // 3. put it inside .hero
```

createElement: a new element, only in JavaScript

className: the class, so the CSS applies

append: inside, after what's there

DATA IN, ELEMENTS OUT

One function, many chips

This afternoon, **createCard(show)** works the same way: give it a show, get back a card.

> Drawing again? replaceChildren() empties the box first.

```
const genresBox = document.querySelector("#genres");

function createChip(text) {
  const chip = document.createElement("button");
  chip.className = "chip";
  chip.textContent = text;
  return chip;
}

function renderChips(list) {
  genresBox.replaceChildren();
  for (const text of list) {
    genresBox.append(createChip(text));
  }
}

renderChips(["Drama", "Comedy", "Crime"]);
```

EVENTS

Listen, then react

Event	Happens when
`click`	something is clicked
`input`	a letter is typed or deleted
`submit`	a form is sent: Enter, or its button

The listener runs **later**, every time the event happens.

```
const searchInput =
  document.querySelector("#search");

searchInput.addEventListener(
  "input",
  () => {
    console.log(searchInput.value);
  }
);
```

THE EVENT OBJECT

Stop what the browser would normally do

A link opens another page. A form reloads the page.

event.preventDefault() cancels it.

```
const homeLink = document.querySelector(".nav a");

homeLink.addEventListener("click", (event) => {
  event.preventDefault();
  console.log("Staying on this page");
});
```

Forms: listen for "submit" on the <form>, not "click" on its button.

STATE → RENDER

Events change the state. render() draws.

event
someone clicks or types

↓

state
one object with everything that can change

↓

render()
draws the page from the state

```
const state = { clicks: 0 };
const heading = document.querySelector("h1");

function render() {
  heading.textContent =
    `Clicked ${state.clicks} times`;
}

heading.addEventListener("click", () => {
  state.clicks = state.clicks + 1;
  render();
});

render();
```

MY LIST

An array of ids, in the state

Ids, not names: unique, and they never change.

filter makes a new array, so store it: picks = picks.filter(…)

```
let picks = [2, 5];

picks.push(7);
// [2, 5, 7]

picks = picks.filter((id) => id !== 5);
// [2, 7]

picks.includes(7);   // true
```

LOCALSTORAGE

Remember My List after a reload

Variables vanish on every reload. localStorage stays.

It keeps **text** only, so turn the array into JSON text and back.

```
// save
localStorage.setItem(
  "picks",
  JSON.stringify(picks)
);

// load, after a reload
const saved = JSON.parse(
  localStorage.getItem("picks")
);
```

THE BIG IDEA

Events change the state. render() draws the page from the state.

THIS AFTERNOON IN THE LAB

Eleven steps, about three hours

Steps 1–2

Elements

1 Find an element

2 Build an element

Steps 3–5

Cards

3 A function that builds a card

4 A card for every show

5 Draw any list of shows

Steps 6–8

Events

6 React to typing

7 Stop the page reloading

8 When nothing matches

Steps 9–11

State, My List

9 Genre buttons and one render()

10 My List

11 Remember My List

Milestone: add two shows to My List, press Run again: they're still there.

STUCK?

In this order

1

Read the message

The “Not yet” box says what to fix.

2

Console

A red line names the problem and the line number.

3

Hint

Two per step. The second is nearly the answer.

4

Answer

Compare it with yours. Find the difference.

5

Ask

Your teacher, or the person next to you.

Most of the time, steps 1 to 3 are enough. That's how developers work too.

END OF THE DAY

Before you go

1. querySelector("#grid") finds nothing: what do you get?
2. Order: render() draws / a chip is clicked / state.genre changes
3. Why JSON.stringify before localStorage.setItem?

FINISHED EARLY?

Extra challenges

Only My List · Best rated first · A details box

TOMORROW

Real shows from the internet

ShowShelf talks to TVmaze: real shows, real search.